

Riley Byers

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EDUCATION

Brandeis University

Bachelor of Arts Computer Science, Bachelor of Arts Psychology, Creative Writing Minor

May 2025

GPA: 3.56/4.00

Coursework: *Perception: Human, Animal, and Machine, Natural Language Processing, Applied Linear Algebra, Research Methods, Data Structures and the Fundamentals of Computing, Discrete Structures, Human Neuropsychology*

SKILLS

Languages: Java • Python • TypeScript • SQL • JavaScript • C# • R • HTML

Frameworks & Libraries: React Native • Expo • Supabase • NumPy • Pandas

Developer Tools: Git • GitHub • Linux • Mapbox • Unity • Blender • Figma

Concepts: REST APIs • Object-Oriented Programming • Data Structures & Algorithms • Backend Development • Database Design • Machine Learning • Data Processing

Additional: Grant Writing • Intermediate Spanish

PUBLICATIONS

Pardo, M. A., L. Gallagher, **R. Byers**, J. M. Winiarski, J. J. Keane, M. Z. Peery, and C. M. Wood. 2026. Passive acoustic monitoring reveals surprising patterns of avian community antipredator behavior at a regional scale. *Ecology*. <https://doi.org/10.1002/ecy.70362>

Huang, L.-Y., **R. Byers**, J. A. Branch, A. Syed, J. E. McDowell, and B. A. Clementz. Interactive ocular motor set-shifting task evoked distinct electrophysiological markers for stages of cognitive flexibility. *Cognitive Neuroscience Society Annual Meeting*, in review [conference abstract].

Zachery, A., F. M. Cornero, L. A. Hartsfield, **R. Byers**, C. Mulligan, and I. M. Pepperberg. Perception of the Ebbinghaus-Titchener illusion in grey parrots (*Psittacus erithacus*): many variables may affect results. *Journal of Comparative Psychology*, in revision (revise and resubmit).

SELECTED EXPERIENCE

Rock Climbing Head Coach/Positive Youth Development Coach – Waltham, MA

Competitive Team Head Coach | Central Rock Gym

August 2021 – Present

- Lead youth climbing programs for athletes ages 7–18, developing structured practices around movement, strength, problem-solving, and competition preparation.
- Coordinate with coaches to evaluate athlete progress and adapt programming across different ages, abilities, and developmental needs.
- Communicate with athletes and families while supporting a safe, inclusive, and growth-oriented team environment.

Rock Climbing Setter – Watertown, MA

Route Setter | Central Rock Gym

October 2025 – Present

- Designed and iteratively refined climbing routes for a diverse user base, balancing creativity, accessibility, and challenge through extensive testing and feedback.
- Collaborated with a high-volume setting team to prototype, evaluate, and improve user experiences while maintaining commercial safety and quality standards.

Cognitive Research Assistant at Ornithology Lab – Sommerville, MA

Research Assistant | Alex Foundation

June 2025 – June 2026

- Manage daily animal care and health monitoring, communicating to lab leadership while supporting safe handling and enrichment.
- Execute study protocols, record subject responses, and maintain accurate, reproducible research documentation.
- Analyze behavioral data in R and collaborate with researchers to select methods, structure datasets, and interpret experimental outcomes.

Computational Research Assistant – Virtual
Research Assistant | University of Georgia

July 2024—July 2024

- Conducted extensive research on R packages for web crawling and gap discovery in research fields, leading to the identification of tools that future students can easily identify and use
- Utilized bibliometric methods to create comprehensive citation networks, including co-occurrence and geographic analyses, which provided information on the value of this package, and its potential future uses
- Developed and organized a detailed guide on the methodologies and tools used, improving the onboarding process for future researchers by outlining the steps involved in bibliometric analysis.

Cornell Lab of Ornithology Internship– Virtual
Undergraduate Research Assistant | Cornell Lab of Ornithology

May 2024 – August 2024

- Improved a large bird call database by using R for data cleaning and optimization, enhancing the efficiency of data access for future machine learning steps
- Used R to create unsupervised classification cluster analysis using feature embeddings in R to categorize bird vocalizations, in order to process large amounts of chickadee data with minimal human hours
- Wrote code in R to identify ratios of chickadee calls from complex datasets to assist in distinguishing bird call patterns
- Identified chickadee call types manually from large audio datasets, aiding in the verification and validation of automated analysis tools.

Tech Internship at Giv – Online
Intern | Giv

February 2024 – April 2024

- Conducted searches to identify missing non-profits for database inclusion, optimizing coverage
- Formulated a structured method for adding new non-profits, reducing processing time from 10 to 5 minutes per entry
- Attended team meetings, accommodating sudden schedule changes and ensuring alignment with fellow interns

University of Georgia Psychology Department Internship – Athens, GA
Research Intern | Clinical and Cognitive Neuroscience Lab

May 2023 – August 2023

- Mastered R programming for eye movement data analysis, transforming raw spreadsheet data into line graphs and computing reaction time and eye velocity, successfully processing and analyzing data for 20 participants
- Performed calibration tests of monitor and eye movement tracker to ensure correct calibration and synchrony with EEG and test paradigm, these tests were used to ensure consistency of data for each participant
- Improved EEG data reliability for subsequent analysis by pre-processing datasets, which included learning the software, filtering out the noise, and accurately marking bad channels

ADDITIONAL PROFESSIONAL EXPERIENCE

Biological Fieldwork — Marine Species Distribution Research

Fieldwork Assistant | Seasonal; 2021 – Present

- Collected and identified invasive porcelain crab specimens in coastal environments to support University of South Carolina research on species distribution, habitat, and climate range.

Summer Camp Counselor – Athens, GA

Head Counselor | Active Climbing

May 2024 – August 2024

- Led games and climbing activities, resolved conflicts, and maintained a structured, supportive environment that balanced camper safety, independence, and exploration.

Front Desk Receptionist– Athens, GA

Receptionist | Active Climbing

January 2022 – August 2022; May 2024 – August 2024

- Managed member check-ins, waivers, sales, equipment setup, facility orientations, and introductory belay instruction while creating a welcoming and safety-focused experience for visitors.

Rock Climbing Setter – Athens, GA

Route Setter | Active Climbing

June 2023 – August 2023

- Collaborated with the setting team to design, test, and refine climbing routes across varied difficulty levels, balancing movement quality, creativity, accessibility, and safety.

Rock Climbing Coach - Athens, GA

Recreational Team Coach | Active Climbing

January 2022– August 2022

- Designed and led practices and conditioning with the goal of introducing young climbers to the sport in a fun and playful way

American Ninja Warrior/Parkour Coach – Watkinsville, GA

Coach for Pre-School Team & Birthday Parties | Oconee Ninja World

October 2019 –February 2020

- Led age-appropriate parkour and obstacle-based activities for preschool classes and birthday parties, demonstrating foundational movement skills and maintaining a safe, engaging environment.

PROJECT EXPERIENCE

AllTrees — Mobile Application

React Native, Expo, Supabase, Mapbox, TypeScript, Git

- Developed a full-stack mobile application for discovering and documenting recreational tree-climbing locations through an interactive map-based interface.
- Designed Supabase data models and authentication workflows to support user accounts, user-generated location data, and future community features.
- Integrated interactive mapping, location search, custom annotations, and reusable interface components with centralized application state.
- Structured the application for future expansion, including filtering, comments, image uploads, safety information, and premium functionality.

Interactive Virtual Reality Maze Game

JavaScript, HTML, A-Frame, Blender

- Built a browser-based virtual reality maze game using A-Frame and JavaScript.
- Implemented collision detection, physics-based interactions, event-driven gameplay, and an AI-controlled enemy pursuit system.
- Modeled and integrated custom 3D assets in Blender and optimized interactions for responsive, real-time gameplay.

Machine Learning Text Classification

Python, NumPy, Natural Language Processing

- Implemented Naïve Bayes and Perceptron classifiers from scratch for supervised text-classification tasks.
- Developed unigram and bigram feature-extraction pipelines and evaluated how different feature configurations affected classification performance.
- Built training and prediction workflows using NumPy-based data structures rather than prebuilt machine-learning libraries.

Scientific Literature Network Analysis

Python, R, Web Crawling, Bibliometric Analysis, Data Visualization

- Researched and evaluated web-crawling, text-mining, and bibliometric tools for identifying relationships and research gaps across scientific literature.
- Developed data-processing workflows to construct citation, keyword co-occurrence, and geographic relationship networks.
- Produced reproducible documentation explaining the selected tools, analytical methods, and implementation process for future researchers.

Charles River Museum Youth Engagement Initiative — DeisHacks

Human-Centered Design, User Research, Grant Research, Team Collaboration

- Collaborated on a four-person team to develop a youth-engagement strategy for the Charles River Museum of Industry and Innovation.
- Conducted need-finding and audience research to propose programming designed to increase engagement among younger visitors.
- Researched potential funding sources and helped draft a grant application supporting the museum's educational and community-outreach goals.
- Received first place and Best Overall Hack at the 2023 DeisHacks Hackathon.

AWARDS & HONORS

DeisHacks Hackathon — Best Overall Hack, First Place

Brandeis University | Spring 2023

- Collaborated on a four-person team to develop a youth-engagement proposal and funding strategy for the Charles River Museum of Industry and Innovation.

World of Work Social Justice Fellowship

Brandeis University | Summer 2023

- Awarded funding to complete a summer internship with an organization addressing issues of social justice.

Brandeis Justice Scholarship

Brandeis University | 2021-2025

- Merit-based scholarship recognizing students' potential contributions to the Brandeis community.

CREATIVE PUBLICATIONS & AWARDS

- **Andrew Grossbardt Memorial Poetry Prize** — First Place | Spring 2024
“In Which Illness Is a Dull Red Thing with Feathers”; published in Laurel Moon
- **Letter Review Competition** — Shortlisted | Spring 2023
“In Which Illness Is a Dull Red Thing with Feathers”; selected as one of six finalists.
- **Antelope Hill Publishing Competition** — Selected for Publication | Summer 2023
“A Gull Calls Me,” published in the Touch Grass collection.

LEADERSHIP & COMMUNITY INVOLVEMENT

Board of Directors — Voices Meets Minds

Treasurer | Remote

June 2026 – Present

- Oversee organizational finances, including bank accounts, expense tracking, reimbursements, budgeting, and financial recordkeeping.
- Support funding acquisition by researching opportunities, contributing to fundraising strategy, and helping plan for the nonprofit’s continued growth.
- Collaborate with organizational leadership to expand a digital mental-health storytelling and education platform encompassing community submissions, social media, a website, and a newsletter.

Brandeis Rock Climbing Club – Waltham, MA

Captain | Spring 2024 – Spring 2025

Undergraduate Representative | 2023

- Developed and led practices, drills, and group discussions for climbers across a range of experience and ability levels.
- Supported route setting and training-program development while promoting club participation and competitive-team cohesion.